ZIN HMUE PAING

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EDUCATION

Temasek Polytechnic — Singapore

Diploma in Computer Engineering; CGPA: 4.0 / 4.0 **(Ranked #1, Common Engineering Programme)** — Apr 2024 – May 2027

- **Coursework:** Python Programming, OOP / Java, Full-Stack Development, AI & Machine Learning — All Distinctions
- **Awards:** Engineering Scholarship (3× recipient), Director's List (Top 10%, 2024/25 & 2025/26), TP LEAD EXCEL Programme (Top 1%)

EXPERIENCE

IT Intern (AI Engineering) — May 2026 – Sep 2026

The Coca-Cola Company — Singapore

- Improved document retrieval accuracy from **71% to 88%** on an AI support assistant used by **190 staff**, by replacing vector-only search with hybrid search and semantic reranking
- Implemented an evaluation harness over **142 labeled tickets** scoring **0.89 groundedness** and **0.91 relevance**, gating every prompt and retrieval change before release
- Shipped the Request and Handoff agents with a mandatory human-approval gate on inventory actions; covered Tier-1 ticket volume fell **38%** post-launch
- Traced three production defects to root cause (degraded OCR, stale inventory data, dead documentation links) and added automated checks to prevent recurrence

Software Engineer Intern — Apr 2026 – May 2026

Datality Lab / Moodie.AI — Singapore

- Developed a job skills portal that loads a **272K-row** SkillsFuture dataset into a client-side SQLite database (sql.js/WebAssembly), delivering **sub-100 ms search** across 50K+ skill-to-role mappings
- Designed a speech analytics pipeline that flagged **˜3,950 vocal anomalies** across 17 hours of audio from 5 ASD students, replacing manual therapist review with automated baseline detection
- Cut audio processing time by **50%** through parallel execution, and implemented longitudinal trend detection across 25 sessions that flagged which students were improving or declining

PROJECTS

AES Side-Channel Analysis | **3rd Place**, NiCE Hack 2026 (NTU EEE) | Python, SCA Toolkit, Asyn2Secure, Arduino Uno — Sep 2026

- Quantified the trace-volume-vs-window-width tradeoff for AES power-trace key recovery across two targets, cutting a wide-window attack from **5,000 traces** (778-sample window) to **250 traces**, and a second target from **2,000 traces** (26-sample window) to **300 traces**
- Recovered a completely unknown AES key from a live Arduino Uno via oscilloscope and EM-probe capture in **382 traces**, after validating the attack pipeline against a known key in **200 traces**
- Completed three bonus cryptanalysis challenges: a Caesar cipher, an XOR key recovery from an encrypted flag, and a timing-attack recovery of a 5-digit code

Lizzie – AI Desktop Automation Agent | Electron, Node.js, Win32 UIA, Whisper ONNX, PowerShell — Jun 2026

- Developed a Windows AI agent with autonomous desktop automation (**13 interaction types, ˜2 s/step**) and a vision-based screen tutor, both with offline voice via bundled Whisper ONNX
- Delivered a **provider-agnostic AI layer** supporting Claude, Gemini, and GPT through a single API contract

KakiLearn AI | Next.js 16, React 19, TypeScript, Tailwind CSS v4, Claude Haiku, Google Cloud TTS — Feb 2026

- Shipped a web app teaching seniors digital banking via interactive simulators, voice-graded quizzes, and AI-generated courses in **4 languages**; tested with **20 seniors**

Career Quest Map | Python, Pygame, LangChain, Azure OpenAI | **Finalist, Top 4 / 40 teams**, NTU Tech4Good Hackathon 2026 — Jan 2026

- Led development of an AI career exploration game built on LangChain LLM orchestration, with role-play personas, decision trees, and response validation; achieved **4.4/5.0 satisfaction** and **100% completion** across 13 sessions with 11 students

TECHNICAL SKILLS

Languages: Python, Java, SQL, HTML/CSS
Frameworks: React, Next.js, Flask, Tailwind CSS, Electron, TensorFlow, scikit-learn, LangChain, LangGraph
AI & Cloud: RAG, Microsoft Azure, Azure OpenAI, Azure AI Foundry, AWS, GCP, Claude/OpenAI/Gemini APIs, Whisper ONNX, Vercel
Developer Tools: Git, GitHub, SQLite, MQTT, InfluxDB, Grafana, Win32 UIA

CERTIFICATIONS & LEADERSHIP

Certs: Microsoft Certified: Azure AI Apps and Agents Developer Associate | Google AI Professional | NVIDIA Deep Learning | CS50x
President and Mentor, Computer Engineering NeTwork (CENT), Temasek Polytechnic — Apr 2025 – Apr 2026
Peer Tutor, Digital Fundamentals & Engineering Mathematics, School of Engineering — Apr 2025 – Feb 2026